Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

COMPUTATIONAL PROBABILITY AND DESIGN OF EXPERIMENTS [MAT 4053]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) A) A computer prints out 1000 numbers consisting of the 10 integers j = 0, 1, …, 9. The number of times j appears is given by n_j = 85, 110, 118, 91, 78, 105, 122, 94, 101, 96. Test the hypothesis that the numbers j are uniformly distributed between 0 and 9, both inclusive, with 5% level of significance. Threshold value is 16.9. (4)

B) Test runs with 6 models of an experimental engine showed that they operated for 24, 28, 21, 23, 32 and 22 minutes with a gallon of fuel. If the probability of a type I error is at the most 0.01, is this evidence against a hypothesis that on the average this engine will operate for at least 29 minutes per gallon of the same fuel. Assume normality. Threshold value is 3.3649 (3)

C) Let $X_1, X_2, \ldots\ldots, X_n$ be a random sample from $U(-\theta,\theta)$. Obtain the maximum likelihood estimate of θ based on the sample values given below. (3)
0.46, 0.38, 0.61, 0.82, 0.59, 0.53, 0.72, 0.44, -0.59, -0.60

2) A) The following observations have been taken sequentially from a distribution having probability function $p_\theta(x) = \begin{cases} \frac{\theta}{4}, x = -1 \\ 1-\theta, x = 0 \\ \frac{3\theta}{4}, x = +1 \end{cases}$

-1, 0, 1, 1, 1, 0, 0, 0, -1, -1, 1, 1, -1, 0, 1, 1, 0, -1, 1, 0, 1, -1, -1, -1, 0, 1, 0, 1, 1, 1, 0, 1. (4)

Using sequential probability ratio test procedure, test the hypothesis

$H_0: \theta = 0.25\ vs\ H_1: \theta = 0.75$. Test at strength $(\alpha = 0.05 \quad \beta = 0.05)$. $R_m = \frac{p(.|H_1)}{p(.|H_0)}$.

The boundary points (A, B) for the SPRTP of strength (α, β) are $A = \frac{1-\beta}{\alpha}$ and $B = \frac{\beta}{1-\alpha}$

B) The following data is collected on two characteristics. Based on this, can you conclude that there is no relation between the habit of cinema going and literacy. Threshold at 5% level of significance is 3.84. (3)

	Cine-goers	Non-cine-goers
Literate	83	57
Illiterate	45	68

C) Three processes A, B and C are tested to see whether their outputs are equivalent. The following observations of output were made.

A: 10, 12, 13, 11, 10, 14, 15; B : 9, 11, 10, 12, 13; C : 11, 10, 15, 14, 12, 13

Carry out the analysis at 5% level of significance and state your conclusions. Threshold value is 3.68. (3)

3) An industrial engineer is investigating the effect of four assembly methods (A, B, C, D) on the assembly time for a colour television component. Four operators are selected for the study. Furthermore, the engineer knows that each assembly method produces such

A) fatigue that the time required for the last assembly may be greater than the time required for the first, regardless of the method. To account for this source of variability, the engineer uses the Latin square design shown below. Analyse the data from this experiment at 5% level of significance and draw appropriate conclusions. Threshold value is 4.76. (4)

Order of Assembly	**Operator**			
	1	**2**	**3**	**4**
1	C=10	D=14	A=7	B=8
2	B=7	C=18	D=11	A=8
3	A=5	B=10	C=11	D=9
4	D=10	A=10	B=12	C=14

B) Write the incidence matrix for the design below having four treatments and six blocks. (3)

Blocks	Treatments	
1	1	2
2	1	3
3	1	4
4	2	3
5	2	4
6	3	4

C) Estimate the missing observations x_1 and x_2 in the following design, taking their initial approximations as zero and perform three iterations. A, B, C, D and E are the makes of the cars. (3)

Drivers	Speed in miles per hour				
	25	35	50	60	70
D_1	B (19.5)	E (21.7)	A (18.1)	D (14.8)	C (13.7)
D_2	D (16.2)	B (19.0)	C (x_1)	A (17.9)	E (17.5)
D_3	A (20.6)	D (16.5)	E (19.5)	C (15.2)	B (x_2)
D_4	E (22.5)	C (18.5)	D (15.7)	B (16.7)	A (16.0)
D_5	C (20.5)	A (19.5)	B (15.6)	E (18.7)	D (12.7)

4) An industrial engineer employed by a beverage bottler is interested in the effects of 2 different types of 32-ounce bottles on the time to deliver 12-bottle cases of the product. The two bottle types are glass and plastic. Two workers are used to perform a task consisting of moving 40 cases

A) of the product 50 feet on a standard type of hand truck and stacking the cases in a display. Four replicates of a 2^2 factorial design are performed, and the times observed are listed in the following table. Analyse the data and draw appropriate conclusions. F(3, 9)=3.86 and F(1,9)=5.12 at 5% level of significance. (4)

Bottle type	Worker			
	1		2	
Glass	5.12	4.89	6.65	6.24
	4.98	5.00	5.49	5.55
Plastic	4.95	4.43	5.28	4.91
	4.27	4.25	4.75	4.71

B) Obtain the Yates' table for the following data based on a 3^3 factorial experiment.

Factor A: Temperature 25°C, 27.5°C, 30°C

Factor B: Catalyst concentration levels 10%, 12.5%, 15%

Factor C: Time 10, 15, 20 minutes

Yield of chemical Product				
B levels	A levels	C levels		
		1	2	3
1	1	4.565	9.95	14.195
	2	12.75	15.365	18.34
	3	16.38	18.8	23.39
2	1	5.21	11.62	14.47
	2	14.825	19.88	22.935
	3	20.37	25.375	29.68
3	1	6.85	12.775	16.65
	2	15.14	19.65	22.9
	3	22.65	27.25	30.52

(3)

C) Based on the layout given below, identify the design. Give proper justification. (3)

Treatment (Catalyst)	Block (Batch of Raw material)			
	1	2	3	4
1	73	74	-	71
2	-	75	67	72
3	73	75	68	-
4	75	-	72	75

5) Complete the ANOVA table.

A)

Source of Variation	Df	SS	MS	F
Treatment		178.17		
Block		192.25		
Error		109.89		
Total	23	480.31		

(4)

B) The objective of an experiment was to determine if there is a significant difference in the average yield of rice when three different fertilisers were used. Based on the data collected, the following operations were performed. Mention the reasons for performing the same.

i. Box plot

ii. Shapiro-Wilk's test

iii. Leven's test.

(3)

C) Based on the ANOVA and Tukey's HSD given below, write your inference considering the level of significance as 5%.

Source of Variation	Df	SS	MSS	F	p-value
Fertilizer	2	6.07	3.03	7.86	0.0000
Error	93	35.80	0.3859		

(3)

Pairs	Difference	Confidence Interval	p-value
2 - 1	0.18	(-0.19, 0.55)	0.4954
3 - 1	0.62	(0.23, 0.97)	0.0006
3 - 2	0.42	(0.05, 0.79)	0.0209

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